

HOLLOMAN AFB, NM, USA

WMO: 747320

Lat: 32.850N

Lon: 106.100W

Elev: 4158

StdP: 12.62

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23002

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.9	22.4	-3.5	5.3	45.9	0.3	6.5	49.1	26.5	54.5	23.1	53.3	5.0	0	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.0	100.3	62.4	98.4	62.7	96.2	62.6	69.0	84.7	68.1	84.3	67.2	84.0	9.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	111.9	71.1	64.1	104.7	71.9	62.7	99.5	72.9	35.6	83.7	34.7	83.8	33.9	83.7	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.9	18.3	11.4	104.6	7.8	2.3	5.7	106.3	1.2	107.6	-3.2	108.9	-8.9	110.5
			9.5	72.3	7.1	3.3	4.4	74.7	0.2	76.6	-3.8	78.5	-9.0	80.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.0	41.3	46.8	53.9	61.3	70.3	80.0	81.5	80.1	73.6	62.2	50.0	41.9
	DBStd	15.79	6.61	7.72	7.03	7.02	6.22	4.90	3.78	3.64	4.92	7.46	7.10	7.23
	HDD50	822	280	133	44	4	0	0	0	0	0	7	87	267
	HDD65	3076	734	510	345	150	25	0	0	0	5	140	451	716
	CDD50	5205	11	43	166	343	631	901	978	934	709	387	86	16
	CDD65	1985	0	0	2	39	191	451	513	469	264	55	1	0
	CDH74	26996	0	15	183	915	3034	6544	6630	5747	3029	868	31	0
	CDH80	13942	0	0	17	243	1400	3858	3745	3086	1343	250	0	0
Wind	WSAvg	8.1	7.1	8.2	8.9	10.2	9.6	9.2	8.2	7.5	7.2	7.3	6.8	6.9
Precipitation	PrecAvg	11.5	0.7	0.6	0.4	0.5	0.6	0.8	1.7	2.0	1.5	1.1	0.6	0.9
	PrecMax	18.7	1.9	2.7	0.9	1.9	3.0	3.8	3.4	7.7	3.9	3.3	1.7	4.0
	PrecMin	6.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.8	0.0	0.0	0.0	0.0
	PrecStd	2.8	0.6	0.6	0.3	0.5	0.7	0.8	0.7	1.5	1.1	0.9	0.5	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.9	76.6	82.8	89.8	97.9	104.7	103.2	100.2	96.5	89.9	77.4	69.3
		MCWB	47.4	49.0	51.4	54.8	58.4	61.5	62.6	64.8	62.6	59.8	52.5	48.1
	2%	DB	64.7	72.2	79.0	85.7	94.2	101.6	100.0	97.9	93.0	86.0	74.2	65.1
		MCWB	45.7	47.4	50.0	53.2	56.8	60.7	63.9	64.8	62.7	58.7	51.2	47.1
	5%	DB	61.2	68.4	75.7	82.6	90.9	98.9	97.8	95.7	90.5	82.5	70.6	61.0
		MCWB	43.5	46.4	48.8	52.0	56.3	60.8	64.3	65.0	61.9	57.0	50.2	44.7
	10%	DB	57.2	64.2	72.6	79.8	88.0	96.3	95.1	93.3	88.0	79.3	66.5	57.2
		MCWB	41.8	44.8	47.6	51.1	55.2	60.3	64.2	64.9	61.6	55.8	48.5	43.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	52.1	54.1	58.4	63.4	68.3	70.4	70.7	68.8	66.9	57.1	51.1
		MCDB	63.7	67.3	74.5	78.4	85.4	84.3	86.4	83.3	78.9	76.5	67.4	64.3
	2%	WB	46.8	49.5	51.9	55.6	60.8	66.4	69.1	69.0	67.5	63.1	54.4	48.6
		MCDB	60.5	66.6	72.8	78.1	83.9	86.1	86.0	84.9	79.4	76.1	65.9	60.9
	5%	WB	44.8	47.7	50.2	53.9	59.1	65.2	68.2	68.1	66.1	60.4	51.9	46.4
		MCDB	58.5	65.6	71.7	76.4	82.6	86.1	84.8	85.2	79.7	75.9	66.1	58.5
	10%	WB	42.7	45.8	48.5	52.1	57.4	64.0	67.3	67.2	64.9	58.6	49.9	44.2
		MCDB	55.4	63.3	69.8	76.5	81.0	85.9	83.9	85.1	79.5	74.3	65.6	55.7
Mean Daily Temperature Range	5% DB	MDBR	28.8	30.0	31.1	31.3	30.9	29.9	26.0	25.8	26.0	28.8	29.3	26.4
		MCDBR	33.5	35.5	35.4	34.7	34.8	33.4	29.5	28.7	29.6	32.0	34.0	30.9
		MCWBR	19.0	18.6	16.7	15.8	14.8	12.6	9.1	8.4	10.4	13.8	17.6	17.3
	5% WB	MCDBR	29.3	31.7	31.8	32.2	30.5	27.8	24.2	24.4	22.6	25.9	27.6	26.4
		MCWBR	17.2	17.2	15.4	15.2	13.1	10.0	7.2	7.8	8.1	12.1	15.4	15.5
Clear-Sky Solar Irradiance	taub	0.249	0.256	0.276	0.305	0.322	0.344	0.384	0.366	0.331	0.287	0.266	0.253	
	taud	2.615	2.590	2.516	2.411	2.420	2.391	2.357	2.377	2.440	2.564	2.570	2.603	
	Ebn at Noon	316	322	320	312	305	297	284	289	296	305	305	308	
	Edh at Noon	24	27	32	37	37	38	39	38	34	28	25	23	
All-Sky Solar Radiation	RadAvg	1124	1439	1859	2279	2485	2478	2210	2049	1843	1528	1202	1007	
	RadStd	79	114	101	101	98	125	116	82	89	122	79	57	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page